

Module 1: Network + Network Fundamentals and building networks

Day 6

Task 1:

Device	Main Function	Ports
Hub	Broadcasts incoming data to all connected devices	4–8 Ethernet (RJ45) ports
Switch	Forwards data only to the specific device it's meant for, using MAC addresses — much more efficient than a hub	4–48 Ethernet (RJ45) ports; managed switches may add a console port
Router	Connects your home network to the internet, assigns IP addresses (DHCP), and directs traffic between networks	1 WAN port (to modem/ISP), 4 LAN Ethernet ports, power port
Firewall	Filters incoming/outgoing traffic to block unauthorized or malicious access	Often built into the router; dedicated firewall appliances have WAN + LAN Ethernet ports
Access Point	Extends Wi-Fi coverage by broadcasting a wireless signal, often connected to a switch/router via Ethernet	1 Ethernet (uplink) port, power port (or PoE), external/internal antennas

Task 2:

Router (back panel):

- 1 **WAN port** — usually a different color, connects to your ISP's modem
- 4 **LAN Ethernet ports** — connect wired devices (PC, smart TV)
- 1 **Power port** — power adapter input
- Antennas (external, if present) — for Wi-Fi broadcast



Switch (front/back panel):

- Multiple identical-looking **Ethernet ports** (numbered) — all function the same way, connecting wired devices
- **Power port**
- (On managed switches only) a small **console port** — used for direct configuration via cable, not for regular data



Access Point:

- 1 **Ethernet/uplink port** — connects back to the router or switch
- **Power port** (or it draws power through the Ethernet cable itself — called PoE, Power over Ethernet)
- **Antennas** — internal or external, responsible for broadcasting the Wi-Fi signal



Task 3:

Example: Instagram and the devices behind it

When you open Instagram at home, your **access point** is what your phone connects to wirelessly, converting your request into a signal the wired network can carry. That request then passes through your **router**, which checks its routing table, assigns/uses your device's IP address, and forwards the request out to the internet through your ISP connection. If your home setup has a wired device too — say a smart TV also streaming — the **switch** built into (or connected to) your router ensures that traffic to each device stays separate and doesn't get broadcast to devices that don't need it.

Task 4:

Diagnosing slow/disconnecting college Wi-Fi

A very common cause in colleges is that a single access point is trying to serve **too many connected devices** at once — APs have a practical capacity limit (both in terms of number of simultaneous connections and wireless channel congestion), so in a crowded lecture hall or hostel, everyone's phones and laptops compete for the same limited airtime, causing slowdowns and dropped connections.

Possible fix: Upgrading (or adding more) access points — especially newer Wi-Fi 6/6E APs, which handle far more simultaneous device connections efficiently and

support better channel management — would reduce congestion per AP and improve both speed and stability. Simply upgrading the router alone usually won't help much here, since the router's job is routing traffic to the internet, not managing the wireless connection itself — the AP is the actual bottleneck in this scenario.